Classification Error Metrics

Kritanta Saha

Assistant Professor
Dept. of Computer Science & Engineering
Sister Nivedita University

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Outline

Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	**TP** *True Positive*	**FN** *False Negative* **Type II Error**
Actual Negative	**FP** *False Positive* **Type I Error**	**TN** *True Negative*

Sensitivity or Recall or True Positive Rate or Hit Rete etc.

	Predicted Positive	Predicted Negative	
Actual Positive	TP *True Positive*	FN *False Negative* **Type II Error**	Sensitivity $\frac{TP}{(TP+FN)}$
Actual Negative	FP *False Positive* **Type I Error**	TN *True Negative*	

Specificity or True Negative Rate or Selectivity etc.

	Predicted Positive	Predicted Negative	
Actual Positive	TP *True Positive*	FN *False Negative* **Type II Error**	Sensitivity $\frac{TP}{(TP+FN)}$
Actual Negative	FP *False Positive* **Type I Error**	TN *True Negative*	Specificity $\frac{TN}{(TN+FP)}$

Precision or Positive predictive Value etc.

	Predicted Positive	Predicted Negative	
Actual Positive	**TP** *True Positive*	**FN** *False Negative* **Type II Error**	Sensitivity $\frac{TP}{(TP+FN)}$
Actual Negative	**FP** *False Positive* **Type I Error**	**TN** *True Negative*	Specificity $\frac{TN}{(TN+FP)}$
	Precision $\frac{TP}{(TP+FP)}$		

Negative Predictive Value or True Negative Accuracy etc

	Predicted Positive	Predicted Negative	
Actual Positive	TP *True Positive*	FN *False Negative* **Type II Error**	Sensitivity $\frac{TP}{(TP+FN)}$
Actual Negative	FP *False Positive* **Type I Error**	TN *True Negative*	Specificity $\frac{TN}{(TN+FP)}$
	Precision $\frac{TP}{(TP+FP)}$	Negative Predictive Value $\frac{TN}{(TN+FN)}$	

Accuracy

	Predicted Positive	Predicted Negative	
Actual Positive	**TP** *True Positive*	**FN** *False Negative* **Type II Error**	**Sensitivity** $\frac{TP}{(TP+FN)}$
Actual Negative	**FP** *False Positive* **Type I Error**	**TN** *True Negative*	**Specificity** $\frac{TN}{(TN+FP)}$
	Precision $\frac{TP}{(TP+FP)}$	**Negative Predictive Value** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

Other Metrices

- False Positive Rate (FPR):

$$\text{FPR} = \frac{FP}{FP + TN} \tag{1}$$

- False Negative Rate (FNR):

$$\text{FNR} = \frac{FN}{FN + TP} \tag{2}$$

- F1-score:

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Question

A hospital is testing a new AI-based diagnostic tool to detect whether a tumor is **malignant** or **benign**. The test was conducted on **500 patients**, and the results are as follows:

- **200 patients have malignant tumors**, among them:
  - **160 tested positive** (True Positives, TP)
  - **40 tested negative** (False Negatives, FN)
- **300 patients have benign tumors**, among them:
  - **240 tested negative** (True Negatives, TN)
  - **60 tested positive** (False Positives, FP)

Using this information, construct the **confusion matrix** and calculate: Sensitivity, Specificity , Accuracy ,Precision ,Negative Predictive Value ,False Positive Rate ,False Negative Rate and F1-score.

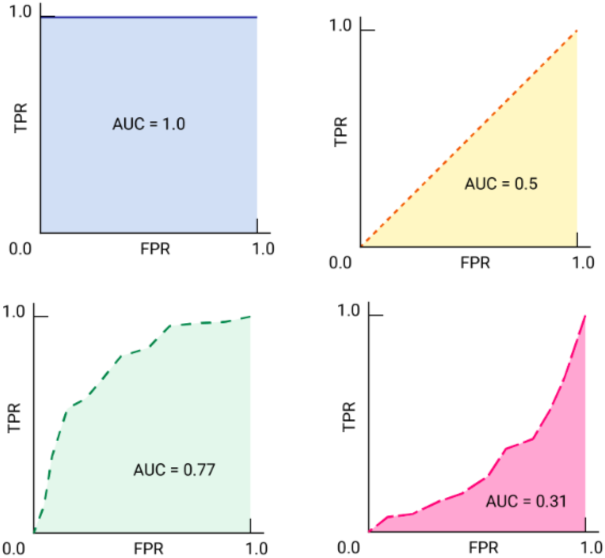
Receiver Operating Characteristics (ROC) Curve

- The ROC curve plots the True Positive Rate (TPR), also known as sensitivity, against the False Positive Rate (FPR), at diverse threshold settings.
- A model with higher TPR and lower FPR is considered better.

Area Under the Curve (AUC)

- It represents the degree of separability.
- The higher the AUC, the better the classifier can distinguish between the positive and negative classes.
- The value of AUC ranges from 0 to 1.
- If AUC=1, The Model can distinguish between the Positive and the Negative class points correctly.
- If AUC=0, The Model with 100% false predictions.
- If AUC=0.5, The model possesses no class separation capacity.

AUC cont..


1.0
TPR
AUC = 1.0
0.0
FPR
1.0
1.0
TPR
AUC = 0.5
0.0
FPR
1.0
1.0
TPR
AUC = 0.77
0.0
FPR
1.0
1.0
TPR
AUC = 0.31
0.0
FPR
1.0

Relation between Sensitivity, Specificity, and Threshold

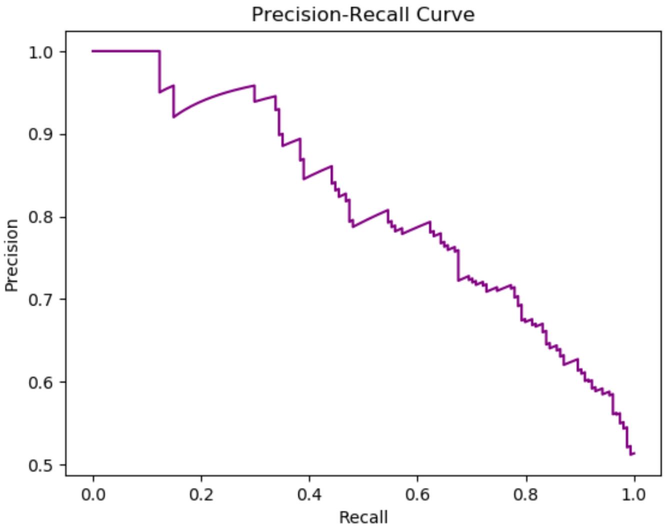
- The Sensitivity and Specificity of a classifier are inversely proportional to each other. So, when Sensitivity increases, Specificity decreases, and vice versa.

- $FPR = (1 - Specificity) \implies$ when Sensitivity (TPR) increases, FPR will also increase, and vice versa.

- Choice of threshold depends on the ability of the AUC-ROC curve to balance between False positives and False negatives.

- Increase in the threshold $\implies$ Increase in positive and negative values $\implies$ increase in the Sensitivity and decreasing in the Specificity.

Precision - Recall Curve



Which approach works best for choosing a classifier?

- **ROC Curve**: Generally better for data with balanced classes.
- **Precision-Recall Curve**: Generally better for data with imbalanced classes.

References

- Book 1: Ethem Alpaydin, “Introduction to Machine Learning”, 3rd Edition.
- Book 2: Tom M. Mitchell, “Machine Learning” , McGraw-Hill
- Online: Machine Learning course by Stanford University.